WENGOLD
OPEN SILICON VERIFICATION

Open-Core Strategy

How an open verification kernel becomes a company

WENGOLD open-sources the validity kernel — the layer that decides which verification results still hold after a change — and sells the scale, governance and support a company needs to run it across a chip program. This document sets out the market, the open-core boundary, the revenue model, the go-to-market path, the competitive field, and the milestones that decide whether it works.

Field	Value
Document	Open-Core Strategy, v0.1
Status	Working draft for discussion
Licence of core	MIT or Apache-2.0
Wedge	Autonomous silicon verification, digital
Primary market	Semiconductor EDA and IP
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Executive summary

Verification is the largest cost in a chip program — 40–50% of the budget on a leading-edge design, and heading toward 60–70% of engineering hours. Most of it is iteration, and the incumbents sell speed, not the thing that removes iteration: a way to know which prior results still hold.

WENGOLD is that layer, built in the open. The kernel, graph and ledger are MIT or Apache-2.0, so a five-person team can verify a design on a free toolchain and a large team can audit every verdict. The company monetises the parts a commercial buyer needs and a hobbyist does not: fleet-scale operation, air-gapped deployment, governance, curated agent packs and support.

~$16–20B
EDA market in 2025, growing 9–13% a year

85%+
held by the Big-3 incumbents

40–50%
of chip budget spent on verification

0
open tools today that answer what still holds

> **The bet.** The kernel is the hard, defensible part — it needs a graph of the whole design and a way to reason about validity, and nobody has open-sourced it. Open the kernel to win the category; sell scale and trust to win the revenue. Incumbent tools cannot follow without rebuilding how their tools store state.

The market: a growing bill for iteration

EDA has grown every year for over a decade, and it grows faster than the semiconductor R&D; it serves — roughly 13% a year against about 7%. The gap comes from exactly where WENGOLD sits: chip complexity, AI accelerator programs, and verification cost at advanced nodes.

Metric	Value	Basis
EDA market size	~$16–20B (2025)	ESD Alliance/SEMI, IDC, third-party estimates; sources differ on scope
Growth	~9–13% CAGR	Multiple 2025–26 market reports
Concentration	Big-3 hold 85%+	Synopsys, Cadence, Siemens EDA
Verification share of chip budget	40–50%, toward 60–70% of hours	Industry practice, leading-edge nodes
EDA growth vs. semiconductor R&D;	+6 points	EDA ~13% vs. R&D; ~7% CAGR

Two forces push buyers toward a new layer. First, cost: as verification approaches two-thirds of engineering hours, the price of rerunning work nobody needed to rerun becomes the dominant line item. Second, access: system companies and hyperscalers now design their own silicon, and the open toolchain (Verilator, Yosys, SymbiYosys, cocotb, OpenROAD) has made the front of the flow reachable without a licence. The layer above it — deciding what to run — has no open option.

Why open core is the right shape here

An open core wins when the free tier creates adoption that the paid tier cannot, and when the paid tier sells something a free user genuinely cannot build alone. Verification satisfies both.

Condition	Why it holds for WENGOLD
The free tier drives adoption	A verification engineer evaluates in a terminal in an afternoon, on their own RTL, with no procurement. The CLI and MCP server are the funnel.

Condition	Why it holds for WENGOLD
Trust requires inspection	Chip teams will not let a closed box decide what is safe to skip. An auditable, open kernel is the only version they will sign off behind.
The paid tier is real	Fleet-scale graphs, air-gapped deployment, SSO/RBAC, audit export and support are operational, not features a community can replicate.
The category is empty	No open tool answers 'what still holds'. The closest work is single-task agents, not a program-level kernel.

Where open source does not fit. We do not open-source the enterprise control plane, the curated agent packs, or the hosted fleet. And we never hold back a capability that decides whether a design is correct — the kernel stays whole in the open core, because trust is the product.

The open-core boundary

The line is drawn at the individual versus the organisation. Everything one engineer needs to verify a design is free and permissively licensed. Everything an organisation needs to run it across many engineers, sites and audits is commercial.

Open core — MIT or Apache-2.0	WENGOLD Enterprise — commercial
Dependency graph, validity kernel, ledger	Hosted graph at fleet scale (millions of nodes)
Agent roles and the MCP server	Curated agent packs and fine-tunes
Open tool adapters and the commercial adapter contract	Air-gapped and on-prem deployment, signed builds
CLI, Python API, CI integration	SSO, RBAC, audit export, policy
Community benchmarks and public evals	Regression-farm orchestration and support SLA

Revenue model

Revenue comes from organisations, not seats, and it grows with the size of the chip program the customer runs on WENGOLD.

Line	What the customer buys	Pricing shape
Enterprise subscription	Air-gapped/on-prem deployment, SSO/RBAC, audit, support SLA	Annual, per design site
Hosted fleet	Managed graph and regression orchestration at scale	Usage + platform fee
Curated agent packs	Domain packs (CPU, NPU, interconnect, security) tuned and validated	Annual add-on
Support and services	Onboarding, methodology, custom adapters	Retainer / project

The community tier stays free forever. We do not meter the kernel, and we do not gate correctness. The commercial pitch is operational risk reduction: a team can run WENGOLD on one machine for free, but not across a fleet, an air-gapped fab, or a regulated audit.

Go-to-market: bottom-up, then enterprise

1. **Land in the terminal.** One command installs the open core; a bundled SoC demo shows the kernel's answer in minutes. No procurement, no sales.

2. **Win the verification lead.** The person who reopens waveforms late at night is the buyer's champion. Give them a result they can show their team.
3. **Prove it on public benchmarks.** Publish evals on open designs so the claim is checkable, not asserted.
4. **Convert on scale.** When the program outgrows one machine — fleet graphs, air-gap, audit — the enterprise tier is the natural next step.
5. **Meet the industry where it gathers.** DAC, Hot Chips and the open-EDA community are where verification leads already are.

> **The funnel metric that matters.** Installs that reach a first kernel verdict on real RTL — not stars, not downloads. A verdict on someone's own design is the moment WENGOLD becomes useful.

Competitive landscape

WENGOLD competes on a different axis from the incumbents: not faster runs, but fewer of them. The field splits into closed platforms, single-task open agents, and the open EDA toolchain underneath.

Player	What they do	How WENGOLD differs
Synopsys, Cadence, Siemens EDA	Full EDA suites; verification is a large line item	They sell tools and speed; the kernel that removes iteration conflicts with how their tools store state
Closed AI-verification startups	Autonomous verification agents (e.g. Tapeout Labs)	Closed and on commercial tools; WENGOLD is open and runs on the free toolchain
Open agent projects	Single-task flows (CoreSmith, UCAgent, VeriPower, xverif, HAVAgents)	They do a task or a pipeline stage; none keeps a validity kernel across revisions
Open EDA toolchain	Verilator, Yosys, SymbiYosys, cocotb, OpenROAD	Complementary: WENGOLD drives them and adds the program-level layer

The moat is the kernel and the graph behind it, not any single agent. A competitor can wrap an LLM around a tool in a weekend; building a graph of the whole design that survives every revision is the hard, compounding part.

Metrics and milestones

Horizon	Milestone	Signal it works
0–6 months	Open core v0.2–0.3, MCP server, public evals	First external team reaches a kernel verdict on their own RTL
6–18 months	v0.4–v0.5, formal loop, regression farm	Open designs verified spec-to-signoff on the free toolchain
18–36 months	First enterprise deployments	Paid pilots in air-gapped environments; a reference customer publishes a result

Risks and mitigations

Risk	Mitigation
Incumbents bundle a similar layer	They would have to change how their tools keep state; the open core wins the community first and sets the interface.
Open core is copied into a closed product	The permissive licence is deliberate: category adoption matters more than exclusivity, and the enterprise tier is operational, not code.
LLM agents are not yet reliable enough	Tools settle every verdict, so agent quality affects speed, not correctness; the kernel is useful with no LLM at all.
Commercial tools close their interfaces	The open toolchain is first-class; commercial adapters are additive, so a closed door slows one path, not the product.
Verification teams resist automation	Agents propose and engineers keep signoff; the kernel sells certainty, not replacement.

Recommendation

1. **Ship the open core as the wedge.** Kernel, graph, ledger and MCP server under MIT or Apache-2.0, with a demo that shows a verdict in minutes.
2. **Publish evals, not claims.** Checkable results on open designs are the only marketing that works with this buyer.
3. **Hold the enterprise line at operations.** Scale, air-gap, governance and support — never correctness.
4. **Measure the first verdict.** Optimise for teams reaching a kernel verdict on their own RTL, then for converting on scale.

> **The one-line thesis.** Open the layer that removes verification iteration; sell the scale and trust to run it across a chip program.

Sources: ESD Alliance / SEMI Electronic Design Market Data (Q4 2025); IDC EDA market shares (2025); SemiAnalysis EDA market primer (2026); vendor and third-party market estimates (2025–26). Verification-share figures are industry practice for leading-edge designs.